Zen Nguyen

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TECHNICAL SKILLS

• **Languages:** Python, Go, TypeScript, JavaScript, SQL
• **AI & Data:** RAG Architecture, Vector Databases (FAISS/ChromaDB), LangChain, OpenAI API, Anthropic API
• **Backend & Systems:** FastAPI, Node.js, Express, RESTful APIs, gRPC
• **Infrastructure & DB:** AWS (EC2), Docker, PostgreSQL, MongoDB, Git

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot) & TitanStore (Raft Database) Jan 2026 – Present
Personal Project · Python, Go, LangChain, FAISS

- Increased job application efficiency by architecting an autonomous AI Co-Pilot in Python that automates end-to-end workflows, utilizing **LLM APIs** to discover and parse thousands of postings.
- Eliminated data hallucinations by building a zero-hallucination RAG engine using LangChain and FAISS, delivering ATS-optimized resumes based on verified user data.
- Guaranteed application state durability and fault tolerance across network failures by developing TitanStore in Go, implementing the Raft consensus protocol and a custom WAL.
- Enabled seamless scaling to 100+ users by designing a high-throughput TCP client API for real-time status updates between Python AI workers and the Go database cluster.

SFU Course Tracker (sfucourseplanner.me) Nov 2025 – Jan 2026
Personal Project · Python, Docker, AWS, FastAPI

- Reduced infrastructure costs and optimized deployments by migrating a full-stack platform from Azure to AWS (EC2) using **Docker** container orchestration.
- Enabled deterministic evaluation of nested boolean prerequisite strings by designing a custom parser that converts unstructured text into an Abstract Syntax Tree (AST).
- Increased data throughput by 10x by engineering **data integrations** via a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting.
- Optimized read performance for complex queries by designing a normalized schema using SQLModel with JSON-type columns to store recursive tree structures.

Gridlock Casino (2D Arcade Engine) Feb 2026 – Present
Collaborative Project · Java, Swing, Maven, JUnit

- Optimized game performance to 60-FPS by architecting a custom 2D grid-based arcade engine in Java using a strict MVC architecture.
- Enhanced enemy AI behavior by implementing BFS for autonomous pathfinding and hitscan vector math for projectile collision.

n54 Present
Personal Project · config, github-config

- Streamlined profile presentation by managing configuration files for GitHub profile.
- Maintained version control of personal branding assets using Git.

EDUCATION

Bachelor of Science, Computing Science May 2025 – Present
Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 – Apr 2025